

4-2-1 Fault Code Listing

Fatal Faults

Fatal faults are conditions that may be unsafe to personnel or the amplifier. If a fatal fault occurs other than in the OFF mode, the amplifier executes an orderly shutdown and returns to the OFF state. The microprocessor prevents the amplifier from again proceeding with its power-up sequence until the internal fault flag is cleared by issuing a "go to off" command (see Chapter 3). Once the red FAULT LED is off (the numerical code will remain displayed), the amplifier may be again commanded into a power-on mode. If the fault condition is still present, the fatal fault sequence will occur again.

Note

"Go To On" Command - Even though the amplifier is on OFF, a "go to on" command must be sent to clear the fault flag.

Non-fatal Faults

Non-fatal faults are used to indicate that an illegal command has been issued or an operational irregularity has been detected which does not warrant complete amplifier shutdown. This level of fault does prohibit an OPERATE mode, but returns the amplifier to STANDBY instead of OFF. The internal flag must be cleared by a "return to standby" command (see the procedure for RF Amp Firmware (20KW)).

Note

"Return To Standby" Command - Even though the amplifier is in STANDBY, a "return to standby" command must be sent to clear the fault flag.

4-2-2 Fault Code Probable Cause

Each fault code is listed below along with a "probable cause" description. This brief description is intended to provide a quick best estimate of where the source of trouble is likely to be located. It is virtually impossible to list all of the possible causes. Service personnel should utilize the information in this Section only as a good starting place.

00: The fault interlock is being held high. The microprocessor is stuck in reset or the cable between the Processor Board and the Front Panel Display Board has an open.

01: The microprocessor detects a "checksum" error. This generally indicates that the address bus is faulted or the EPROM contains incorrect data; definitely a Processor Board problem.

02: Dual port RAM apparently has failed; a Processor Board failure.

03: A/D samples generated on the Processor Board have failed; a Processor Board failure.

04: A/D multiplexer apparently has failed. Either the Processor Board has a problem, or possibly one of the 32 analog convertible signals is unusually errant.

05: A/D on-board self-test has failed; a Processor Board failure.

06: Stepping motor coil is apparently open; very possibly an open in the Motor Cavity Cable Harness or connector; less possible cause is a component malfunction on the Processor Board.

07: Illegal opcode trap; a Processor Board failure.

08: External interrupt high does not cause interrupt; a Processor Board failure.

09: External interrupt low does not cause interrupt; a Processor Board failure.

10: Heat sink (Solid State Amplifier) monitor temperature apparently too hot or cold (very unlikely); probably a failed thermistor or loose screw next to thermistor; perhaps a cable problem or Processor Board A/D problem.

11: SCM cable interlock invalid; either the strobe width is too wide, the cable is open, the strobe signal is high impedance, or most likely, the Processor Board has a problem.

12: +100v supply unacceptable; problem could be on the Power Control Board, the Low Voltage Transformer, an AC Switching Module fuse or relay, a bad cable, or a component failure causing excessive loading (including tubes).

13: +3KV supply unacceptable; failure could be on the High Voltage Rectifier/Filter Board (bad capacitors, diode etc.), the 4.5KVA HV Transformer (unlikely), or possibly a bad IPA tube or 7.5 KV capacitor drawing excessive current.

14: IPA tube heater current unacceptable; most likely a bad IPA tube or Power Control Board circuit. Also cabling including feed-through capacitor is highly suspect.

15: -5 volt supply unacceptable; most likely on the Processor Board. Possibly a bad op-amp on one of the other boards which uses the -5 volt supply.

16: +48 volt supply unacceptable; most likely a bad FET circuit on the Solid State Amplifier Board; possibly a bad Power Control Board circuit or cabling problem.

17: +6KV supply unacceptable is identical to the +3KV supply unacceptable except for the PA tube being a possible cause. Most likely a HV Rectifier/Filter Board problem.

18: PA tube heater current unacceptable; most likely located on the Power Control Board, but very possibly a tube problem. Cabling and feed-through capacitors are suspect.

19: +32 volt supply unacceptable; most probable sources of trouble are the AC Switching Module (fuses or transformer T1), the Power Control Board, Motors, and cabling.

20: +2.5 volt reference unacceptable; most likely a failure on the Processor Board or the AC Switching Module transformer T1 or fuses.

21: +24 volt supply unacceptable; this may arise from any of the supply's numerous loads: Solid State Amplifier, Power Control Board, Processor Board, RF Monitor Module, AC Switching Module, and PA Input Board.

22-25: Motor 1, 2, 3, or 4 zero sensor fault; this either indicates motor is stuck at zero, its optical sensor is stuck open, AFT forced the motor into the zero stop, or cabling has failed.

26-29: Motor 1, 2, 3, or 4 zero not found; possibly a problem with motor driver chips on the Processor Board. Most likely a failed motor assembly: zero sensor, stuck washers, or cabling.

30: General motor fault; one or more of the four motors suddenly (by the microprocessor's perception) no longer has the correct number of steps or has a changed zero position. This most likely indicates a motor assembly failure (usually stuck counter washers); possibly a driver failure on the Processor Board.

31: Unused.

32: UNBLANK longer than 20 milliseconds; either the UNBLANK signal generated by external equipment is too long, or the Processor Board has a problem.

33-34: Unused.

35-38: FET 1, 2, 3, or 4 coarse convergence failure; a problem with either the Solid State Amplifier Board (most likely), the Processor Board, or cabling in between.

39-41: Unused.

42-45: FET 1, 2, 3, or 4 fine convergence failure; identical to faults 35-38 probable cause.

46-49: FET 1, 2, 3, or 4 tracked out of adjustable range; this problem is identical in probable cause to faults 35-38 except that an even higher probability exists for the failure to exist on the Solid State Amplifier Board.

50: +100 volt supply soft-start failure; this early failure of the +100 volt supply is probably due to an excessive load by one of the tubes or a circuitry failure on the Power Control Board. It is also very possible that a problem exists in the AC Switching Module relays or fuses; perhaps a failed Low Voltage Transformer (not very likely).

51: +100 volt supply run failure; identical to fault 50 probable cause.

52: IPA tube heater current unacceptable at start; identical to fault 14 probable cause.

53: PA heater current unacceptable at start; identical to fault 18 probable cause.

54: Unused.

55: +3KV supply unacceptable immediately after soft-start; this failure is most likely due to a problem in the High Voltage Rectifier/Filter Board, and very possibly a relay problem in the AC Switching Module. Other sources of the failure could be the 4.5KVA HV Transformer, loading in the VTAC, and bad A/D on the Processor Board.

56: +6KV supply unacceptable immediately after soft-start; identical to fault 55 probable cause.

57: +3KV soft-start failure; differs from fault 55 only in timing.

58: +6KV soft-start failure; identical to fault 56 except for timing.

59: +3KV supply or +6KV supply run failure; shares the same probable causes as faults 55 through 58.

60: IPA AFT does not converge; this indicates that sufficient power exists for AFT but the motors cannot force a zero-phase condition. Problems could exist with the motors, motor drivers, IPA phase detector, Processor Board A/D, IPA tube, or cabling.

61: PA AFT does not converge; identical to fault 60 probable cause.

62: Insufficient power reaching tube output for AFT; most likely this problem is due the Solid State Amplifier oscillator or path switching. Also very likely is a bad connection between Solid State Amplifier and the VTAC. Other potential sources of failure could be the IPA tube, IPA input board, IPA phase detector, Processor Board A/D, and cabling.

63-69: Unused.

70: Change mode command while in operate; if no such command is being issued by the external controlling computer, then the Processor Board is definitely at fault.

71: Frequency command fault; if the frequency command byte is correct, then the Processor Board is the failure.

72: Invalid mode command; identical to fault 70 probable cause.

73: BODY mode forward power too high; if the RF input power to the amplifier is normal, then the failure is most likely in the RF Monitor Module. Possibly the Processor Board A/D has a problem.

74: HEAD mode forward power too high; identical to fault 73 probable cause.

75: BODY mode reflected power too high; if the amplifier load is normal, then the probable cause is the same as for fault 73.

76: HEAD mode reflected power too high; identical to fault 75 probable cause.

77: PA tube grid current too high; most likely an indication that the YC-156 tube is failing. Possible problems on the Power Control Board or Processor Board.

78: IPA tube grid current too high; identical to fault 77 probable cause.

79: PA tube plate current too high; the probable causes are the same as for fault 77 except that the High Voltage Rectifier/Filter Board is also suspect.

80: IPA tube plate current too high; identical to fault 79 probable cause.

81: IPA tube or +3KV arc; a genuine arc will usually leave a visible mark, and will most probably eventually occur again. Tube arcs may occur internally and will not be visible; if these persist, the tube must be replaced. It is also possible that the arc detect circuitry on the Processor Board is giving false indications.

82: Cover interlock 1 (on RF Monitor Module) open; if the Motor Cavity Right Side Cover is in place, then the switch has failed, a cable is open, or Processor Board has failed.

83: Cover interlock 2 (on AC Switching Module) open; identical to fault 82.

84: RF MON cable disconnected; if RF MON cable is correct, then failure is on the Processor Board.

85: Externally controlled +12 volt safety relay not powered; if the RF MON cable is correctly supplying the voltage, then the Processor Board has a problem. (A GE power monitor fault will produce this error code.)

86: PA tube or +6KV arc; identical to fault 81, but with more punch.

87: Arc detect circuit failure; a Processor Board failure.

88: Fan air pressure insufficient; if the fans are turning and the covers are on, then either the pressure transducer has failed, cabling has failed, or an A/D problem exists on the Processor Board (unlikely).

89: Tube air pressure insufficient; identical to fault 88 probable cause.

90: IPA tube idle DC bias current unacceptable; either the tube or the Power Control Board has failed. Possibly a cable, IPA Input Board, or Processor Board A/D problem.

91: PA tube idle DC bias current unacceptable; identical to fault 90.

92: Line voltage too low; if line voltage is always greater than 180 volts, then Processor Board or AC Switching Module has a problem.

93: Microprocessor "Watchdog" failed to fire in time; a Processor Board failure.

94: Microprocessor "Watchdog" intercepted program; a Processor Board failure.

95-99: Unused.